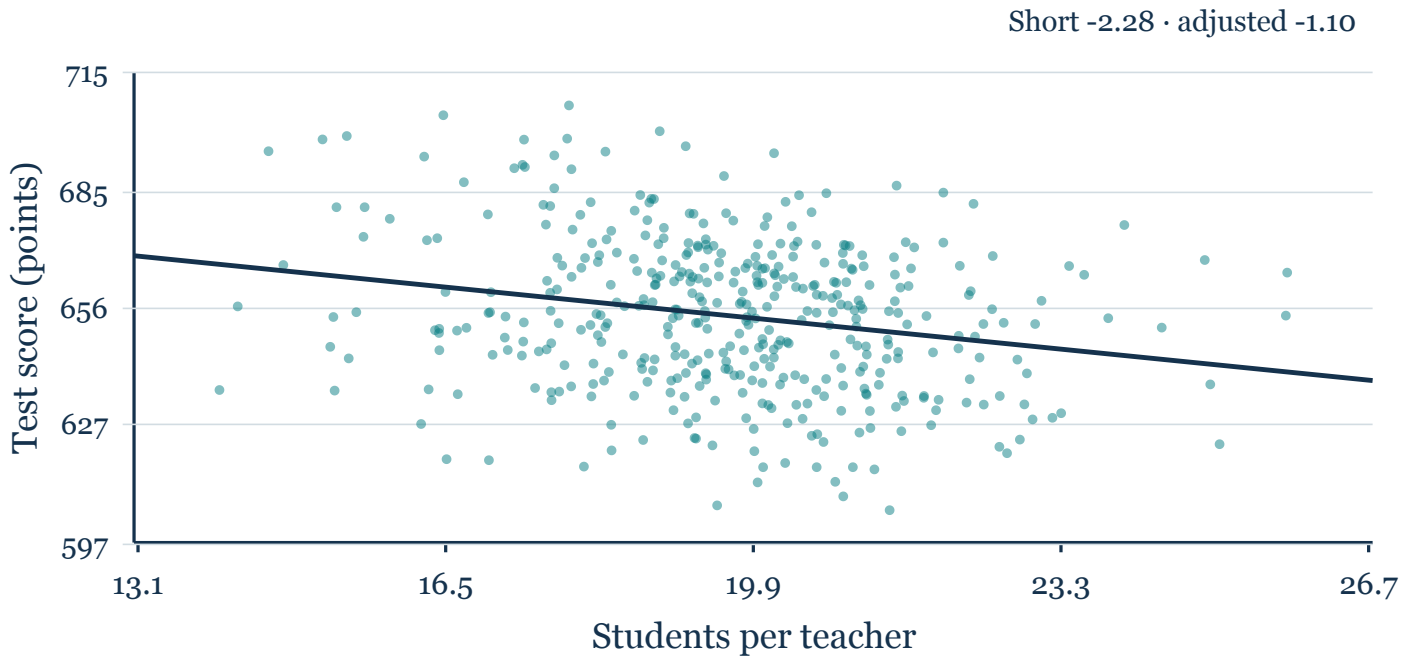
What is hidden in the staffing comparison?

Use a second variable to understand a changed slope.



Observed school data. The control moves an illustrative slope between the two fitted endpoints; intermediate slopes are not new regressions.

Recall the difference between an error and a residual

A property of a fitted line is not a test of the design.

> $e_i = Y_i - \widehat{Y}_i$

> $\sum_i e_i = 0$ with an OLS intercept

Write the longer relationship first

Define what the short model leaves out.

> $Y_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i + u_i$

> Exogeneity for this mean model: $E(u_i \mid X_i, Z_i) = 0$

Two relationships determine omitted-variable bias

Multiply the omitted effect by the omitted-variable association.

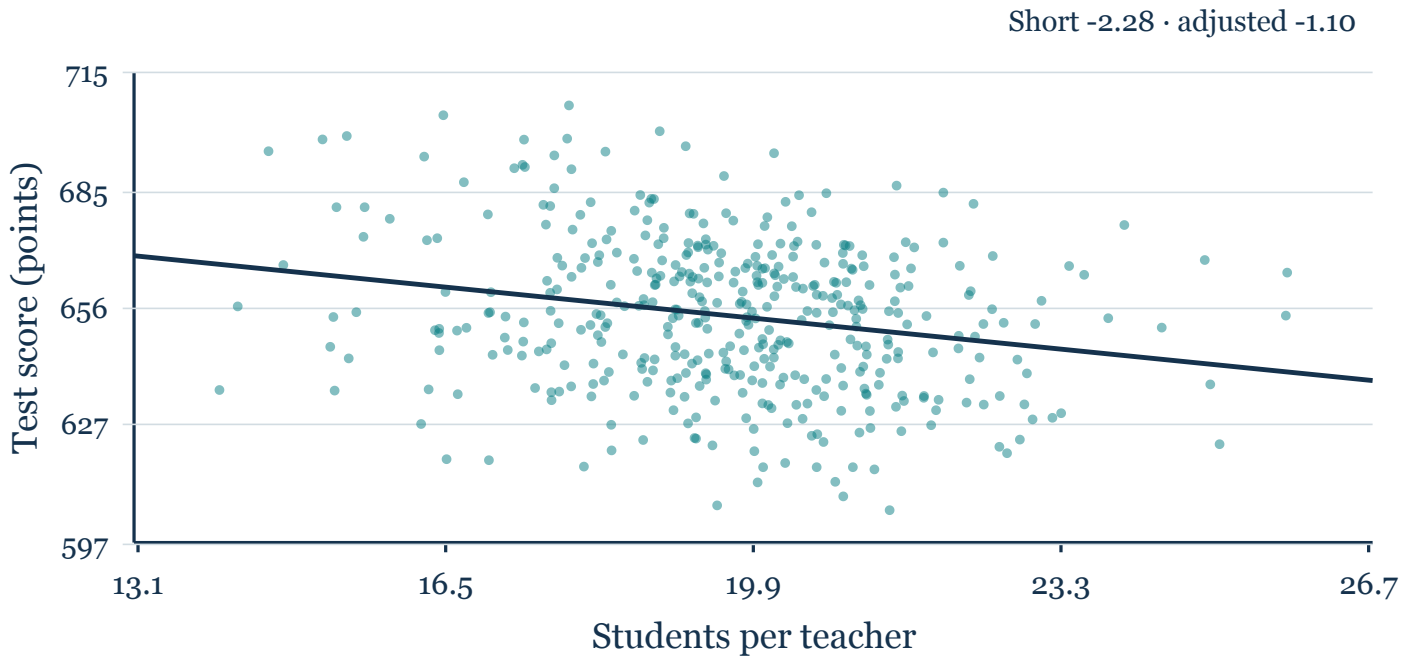
> $Z = \delta_0 + \delta_1 X + v$

> $Y = (\beta_0 + \beta_2 \delta_0) + (\beta_1 + \beta_2 \delta_1) X + (\beta_2 v + u)$

> Short slope minus longer slope: $\beta_2 \delta_1$

Predict the direction before seeing the estimate

The sign depends on both parts of the product.



Observed school data. The control moves an illustrative slope between the two fitted endpoints; intermediate slopes are not new regressions.

Predict omitted-variable bias

Try the comparison before opening the explanation.

Practice

In a wage model, the causal return to schooling is β. Ability raises wages by $\gamma > 0$, and the auxiliary regression of ability on schooling has slope $\delta > 0$. Under the scalar model, what is the omitted-variable bias?

Predict omitted-variable bias: explanation

Follow the calculation and interpret its units.

> The short-regression slope is $\beta + \gamma\delta$, so its bias relative to β is positive. Schooling picks up some of the association between ability and wages. This requires the stated linear model and exogeneity of the remaining error.

Check the actual coefficient movement

Use the same districts in both regressions.

```
. quietly regress score str el

. matrix list e(b), format(%10.6f)

e(b)[1,3]
           str          el       _cons
y1   -1.101296   -0.649777  686.032250
```

Actual Stata coefficient-only display; reproduce with coefficient_workshop.do.

Verify the omitted-variable calculation

An exact sample identity can be checked independently.

```
. quietly regress el str

. matrix list e(b), format(%10.6f)

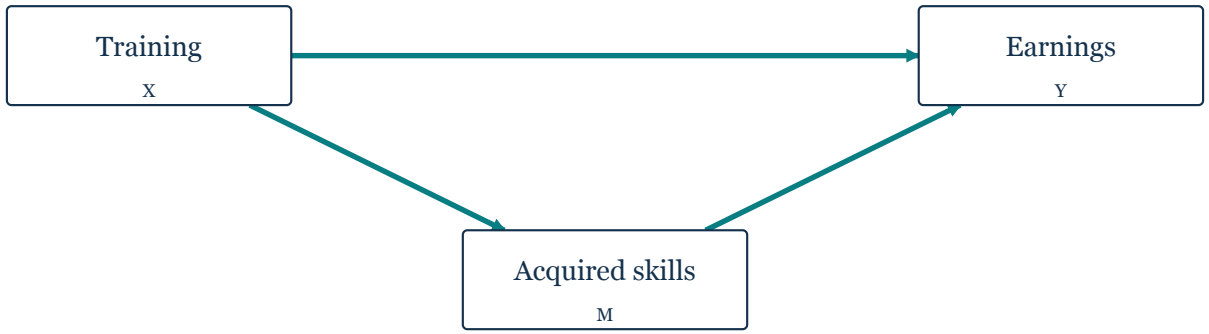
e(b)[1,2]
           str       _cons
y1    1.813719  -19.854055
```

Actual Stata coefficient-only display; reproduce with coefficient_workshop.do.

$$b_{1,\text{short}} - b_{1,\text{long}} = b_2\hat{\delta}_1$$

A confounder and a mediator do different jobs

The timing of a control changes its meaning.



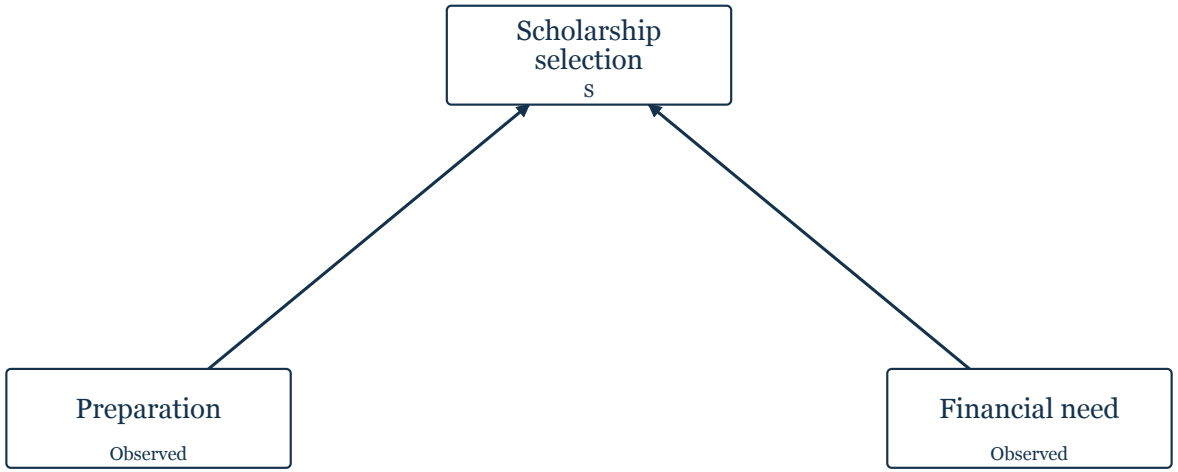
2 open directed routes · 0 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

- Confounder: a prior common influence
- Mediator: part of the effect pathway

Why a collider can be a bad control

Selection into a shared consequence can create a relationship.



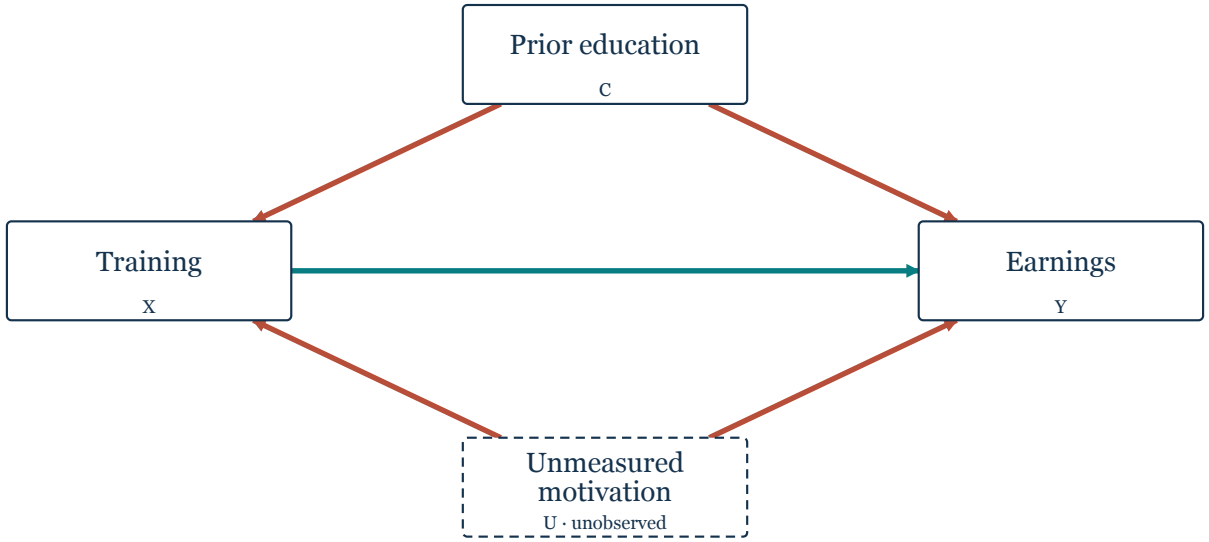
0 open directed routes · 0 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

> Preparation → Selection ← Need

Choose controls for the research question

State what each variable is meant to address.



1 open directed route · 2 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

- New application: training → earnings.
- Prior education is a proposed common cause; motivation remains unmeasured.

Adjustment changes a comparison; it does not guarantee identification

A richer model can still leave an important factor out.

> Identification: do the data and assumptions determine the target?

> Estimation: use the sample to calculate it.

Confounder or mediator?

Try the comparison before opening the explanation.

Practice

Training changes skills, and skills change earnings. The target is training's total effect. Should post-training skill automatically be included as a control?

Confounder or mediator?: explanation

Follow the calculation and interpret its units.

> No. Skill is a mediator on a causal path from training to earnings. Holding it fixed changes the estimand and may introduce other bias. A direct-effect interpretation needs additional assumptions, not merely an added control.

Add income and inspect what changes

An additional observed factor can substantially alter a slope.

```
. quietly regress score str el income

. matrix list e(b), format(%10.6f)

e(b)[1,4]
           str          el      income       _cons
y1   -0.068776   -0.488267    1.494517  640.315497
```

Actual Stata coefficient-only display; reproduce with coefficient_workshop.do.

Keep the estimation sample fixed

Otherwise a coefficient change can have two explanations.

> Compare identical observations

> Then investigate why observations are missing

Keep the sample fixed

Try the comparison before opening the explanation.

Practice

A coefficient changes after adding income, but income is missing for 20% of observations. Why can you not attribute the entire change to controlling for income?

Keep the sample fixed: explanation

Follow the calculation and interpret its units.

> The comparison changes both the regressors and the estimation sample. Refit the short and long models on the same complete-case sample to isolate the specification change. Then consider whether missingness itself creates selection bias.

Interpret a control coefficient cautiously

The variable may be useful without having a causal interpretation.

> The coefficient on English-learner share is an adjusted district association.

> The staffing study may include this variable to address a possible source of confounding, without trying to identify the causal effect of language composition itself.

Fit and causal validity are different properties

Explaining more variation need not improve the comparison.

$$R^2 = 1 - \frac{\sum_i e_i^2}{\sum_i (Y_i - \overline{Y})^2}$$

A useful control plan fits in a short table

Make the assumptions inspectable.

> For each proposed control, write a sentence about timing and a sentence about purpose.

> This exercise reveals disagreements about the economic model before they become disagreements about software.

Use the design to decide what data to collect next

More variables are useful when they address a named problem.

> If earlier learning difficulties drive teacher recruitment, prior outcome measurements may be useful.

> These suggestions follow from different explanations of the error term.

Write the result as an empirical argument

A coefficient is one part of the explanation.

> A useful report starts with the staffing question and identifies a district as the observation.

> Avoid saying that the model 'controls for everything'.

Check the two requirements for omitted-variable bias

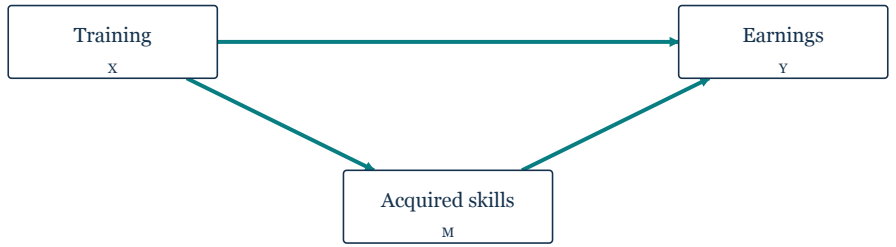
A left-out factor must matter in both relationships.

> If Z does not affect Y in the longer model, its omitted-effect coefficient is zero and the product vanishes.

> Explain these two cases using an economic example.

DAG practice: preserve the intended effect

Trace the routes before choosing a control.



2 open directed routes · 0 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

Practice

Training can affect earnings directly and through acquired skills. The target is the total effect of training. Use the diagram to explain what conditioning on acquired skills removes. Does the graph alone guarantee that an adjusted regression identifies a direct effect?

DAG practice: preserve the intended effect: explanation

Explain what conditioning changes in this stated graph.

> Conditioning on skills closes the route $X \to M \to Y$, removing a mechanism included in the total-effect question. The direct route $X \to Y$ remains in the diagram. A skills-held-fixed comparison therefore concerns a different target. A direct-effect estimate also needs suitable assumptions about treatment assignment, mediator–outcome confounding, intervention meaning, support and the fitted model. Adding a post-training mediator is not, by itself, evidence that these conditions hold. Start with the intended effect, then justify the control choice.

Explain an observed omitted-variable contribution

Reproduce the evidence, then explain its economic meaning.

Practice

Run meeting_02.do using schools.csv. Record the staffing slopes in the short model and the model with English-learner share. Multiply the longer-model language coefficient by the auxiliary staffing–language slope. Check that this product equals the staffing-coefficient movement, then explain why that identity does not identify a causal language effect.

Explain an observed omitted-variable contribution: explanation

A reproducible result needs both a calculation and an interpretation.

> The short staffing slope is -2.279808 and the language-adjusted slope is -1.101296. Their difference is -1.178512. The product of the language coefficient -0.649777 and auxiliary slope 1.813719 gives the same difference before rounding. The identity describes compatible least-squares fits on the same 420 observations. The negative language association and positive staffing–language association make the short staffing slope more negative. Neither component is automatically causal: language share can stand for measured district composition while other background differences remain unobserved. The numerical identity verifies the regression explanation, while the causal interpretation still requires substantive assumptions.

From adjustment to partialling out

Next, make the adjusted comparison visible.

> Today we connected a changed staffing coefficient with a specific omitted-variable calculation.

> The next lesson asks how multiple regression computes the staffing slope after accounting for other variables.